

STAT Project: Marvel Movie Performance Analysis

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Introduction

The goal of this project is to analyze box office performance across the Marvel film universe and explore how different factors predict financial success. The project is organized into two main parts: an exploratory data analysis (EDA) of the dataset and an inferential analysis addressing three research questions using appropriate statistical methods.

The dataset combines two raw sources:

- A Marvel movies financial dataset containing titles, distributor, release dates, reported budgets, domestic and international gross earnings.
- A reviews dataset containing Rotten Tomatoes, Metacritic, and CinemaScore ratings for each movie.

The Marvel movie financial dataset and the reviews dataset used in this project were obtained from a public Kaggle dataset: <https://www.kaggle.com/datasets/minisam/marvel-movie-dataset>.

Both sources were merged using a cleaned movie-title key. Missing values, formatting issues (such as embedded symbols, commas, and text noise), and inconsistent title conventions were cleaned to prepare the dataset for statistical analysis.

After cleaning, the final dataset contains 67 Marvel films with complete box office and rating information.

Part 1: Exploratory Data Analysis

Summary Statistics

We summarize the main financial variables (in millions USD) below:

- **Budget:** Mean ≈ 147.5 million; Median ≈ 150 million.
- **Worldwide Gross:** Mean ≈ 609 million; Median ≈ 502 million.
- **Rotten Tomatoes Scores:** Wide range from low 20s to 100.
- **Release Years:** Movies span earlier pre-MCU films through the current MCU era.

Distribution of Worldwide Gross

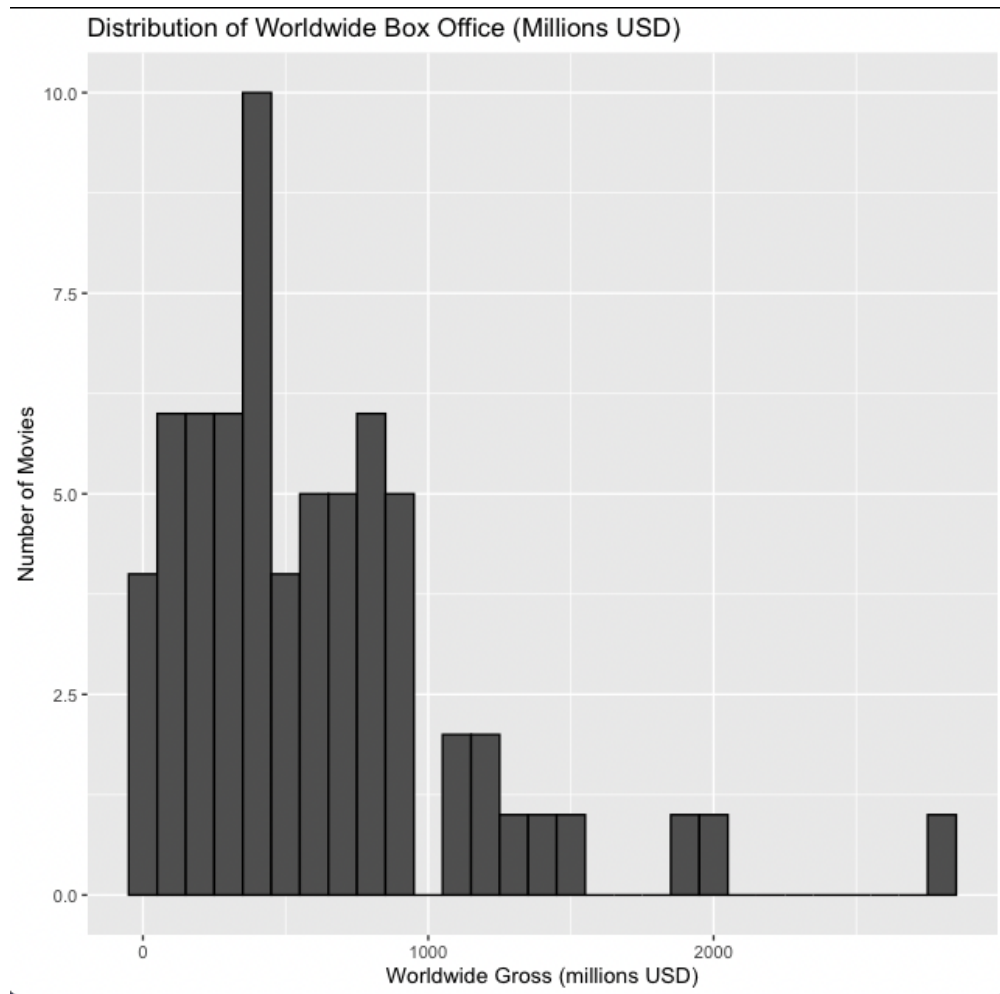


Figure 1: Distribution of Worldwide Box Office (Millions USD)

The distribution is strongly right-skewed, indicating a few major blockbusters earning over \$1B.

Distribution of Budgets

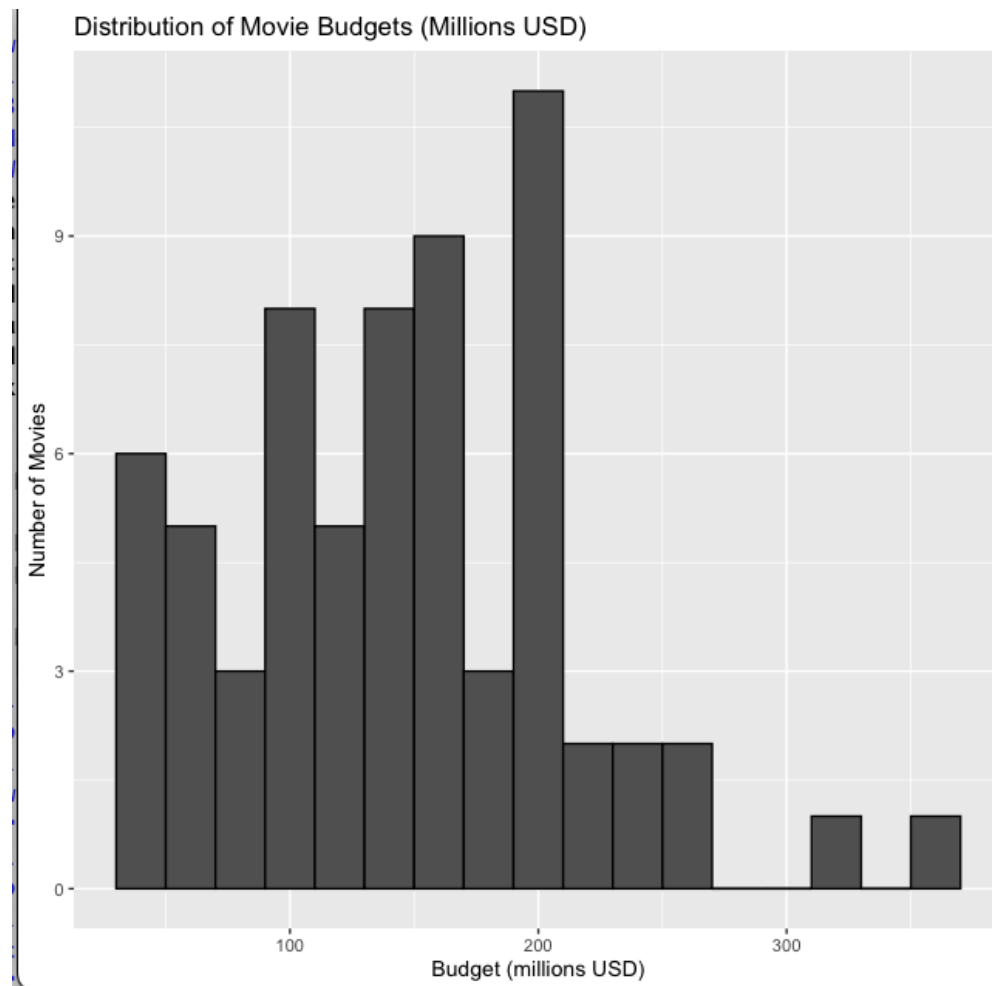


Figure 2: Distribution of Movie Budgets (Millions USD)

Budgets cluster between \$100M and \$200M, with a few very high-budget films.

Budget vs Worldwide Gross

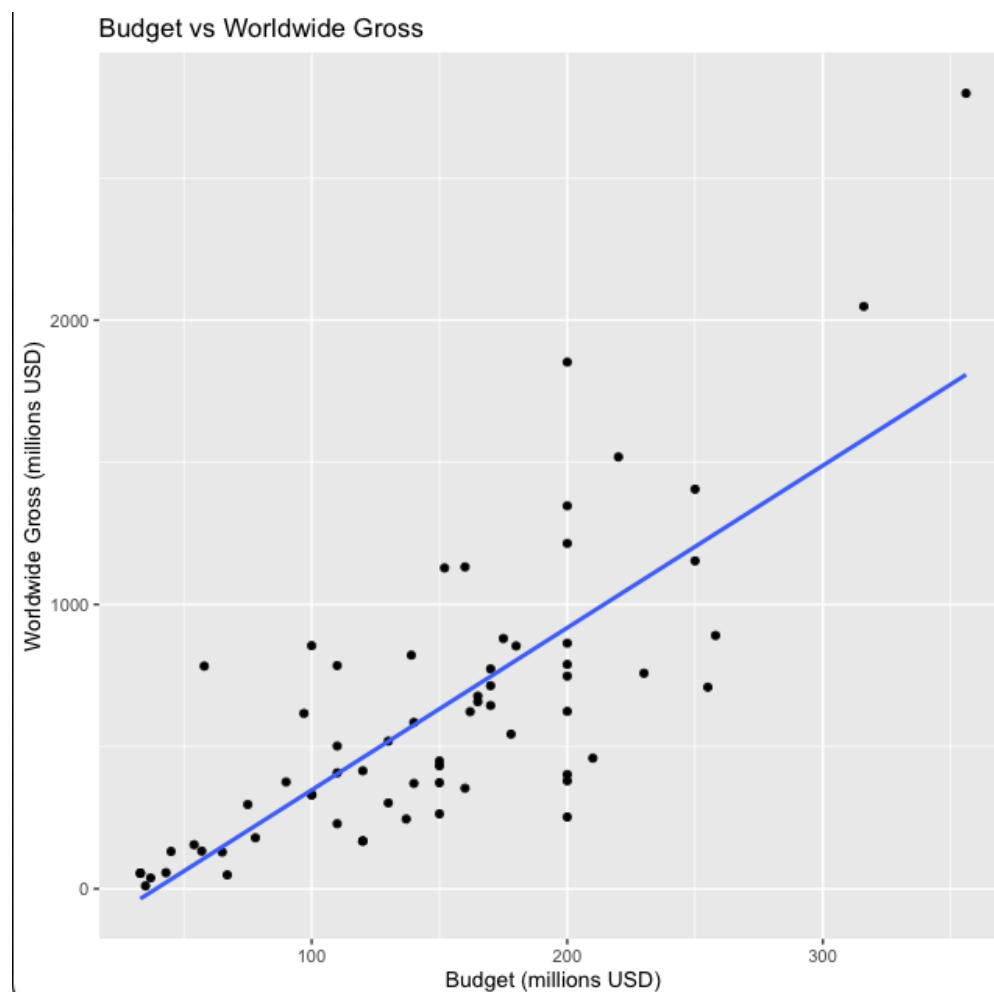


Figure 3: Relationship Between Budget and Worldwide Gross

There is a clear positive association: movies with higher budgets tend to earn more world-wide.

Part 2: Inferential Analysis

This section answers three research questions using statistical inference.

RQ1: Do Marvel movies released in or after 2008 earn significantly more than pre-2008 movies?

To reduce skew, we analyze the log-transformed worldwide gross. Before running the test, I first look at how revenues differ by era and whether the log revenues look roughly normal.

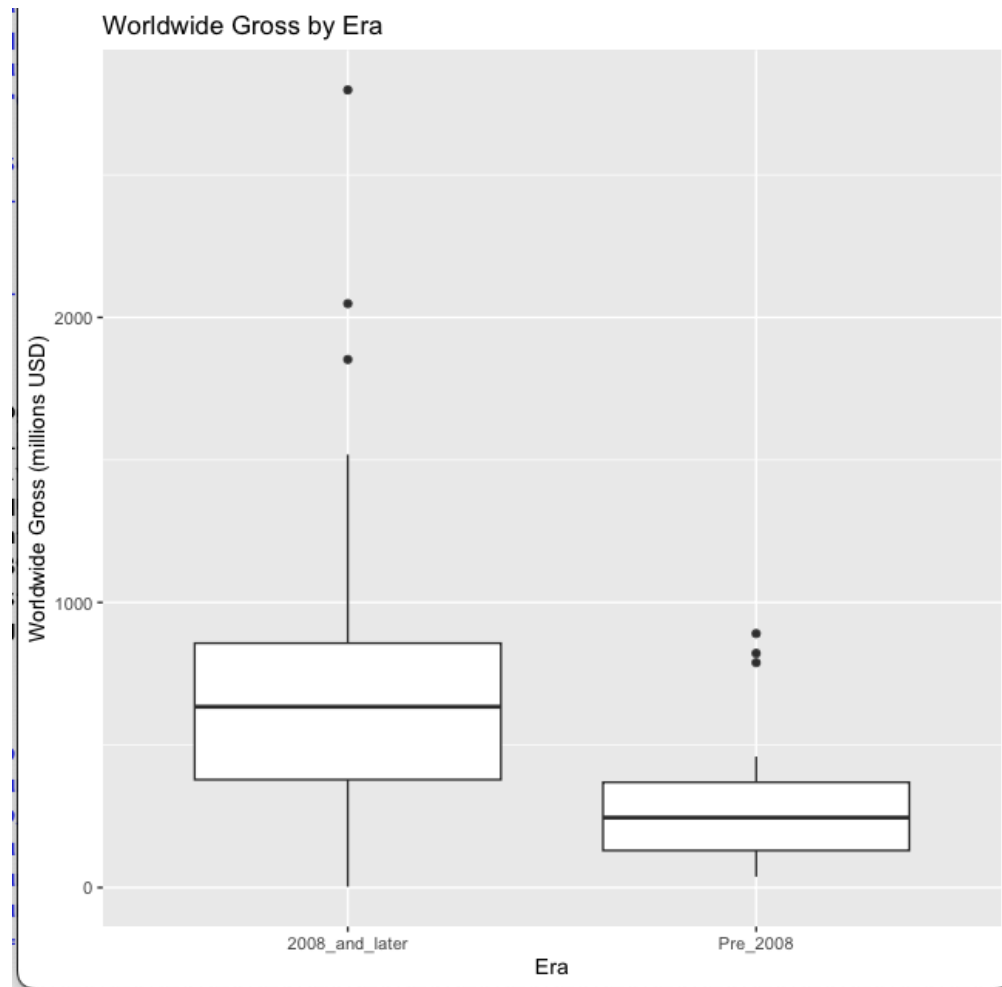


Figure 4: Worldwide Gross by Era (Pre-2008 vs 2008+). The 2008+ movies tend to earn more.

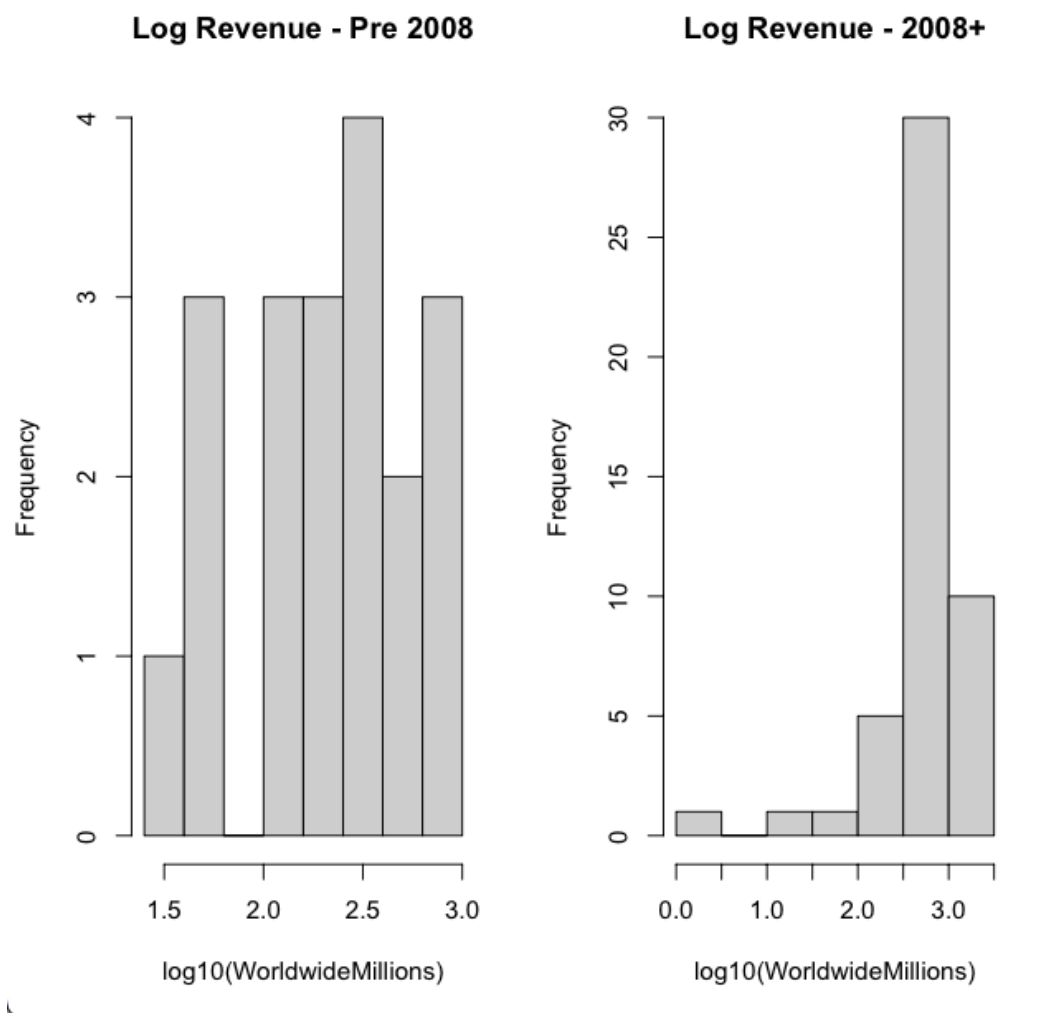


Figure 5: Log-transformed worldwide revenue for Pre-2008 films (left) and 2008+ films (right). Both groups look reasonably unimodal and symmetric after the log transformation, which supports using a two-sample t-test on log revenue.

Two-Sample t-test (Welch)

$$H_0 : \mu_{\log, 2008+} = \mu_{\log, \text{pre-2008}} \quad H_A : \mu_{\log, 2008+} \neq \mu_{\log, \text{pre-2008}}$$

Test results:

$$t = 3.00, \quad p = 0.0045$$

A two-sample Welch t-test is appropriate here because we are comparing the mean (log-transformed) worldwide revenue between two independent groups of movies—those released before 2008 and those released after 2008. Group means:

- 2008+ mean revenue: \$727M

- Pre-2008 mean revenue: \$311M

Conclusion: Movies released in or after 2008 earn significantly more worldwide revenue. This supports the idea that the MCU era boosted overall financial performance.

RQ2: Does average worldwide revenue differ across distributor groups (Disney, Fox, Sony, Other)?

A one-way ANOVA was conducted using log-transformed worldwide gross. A one-way ANOVA is appropriate because we are comparing the mean (log-transformed) worldwide revenue across more than two independent distributor groups, and we want to test whether at least one group differs from the others.

$$H_0 : \mu_{\text{Disney}} = \mu_{\text{Fox}} = \mu_{\text{Sony}} = \mu_{\text{Other}}$$

ANOVA Results

$$F(3, 63) = 15.44, \quad p < 1.3 \times 10^{-7}$$

Conclusion: There is strong evidence that mean worldwide revenue differs across distributors.

Post-Hoc Tukey Test

Key findings:

- Disney movies significantly outperform both Fox and “Other”.
- Sony is significantly higher than “Other”.
- Sony and Fox do not significantly differ.

This aligns with Disney’s role in producing major MCU blockbusters.

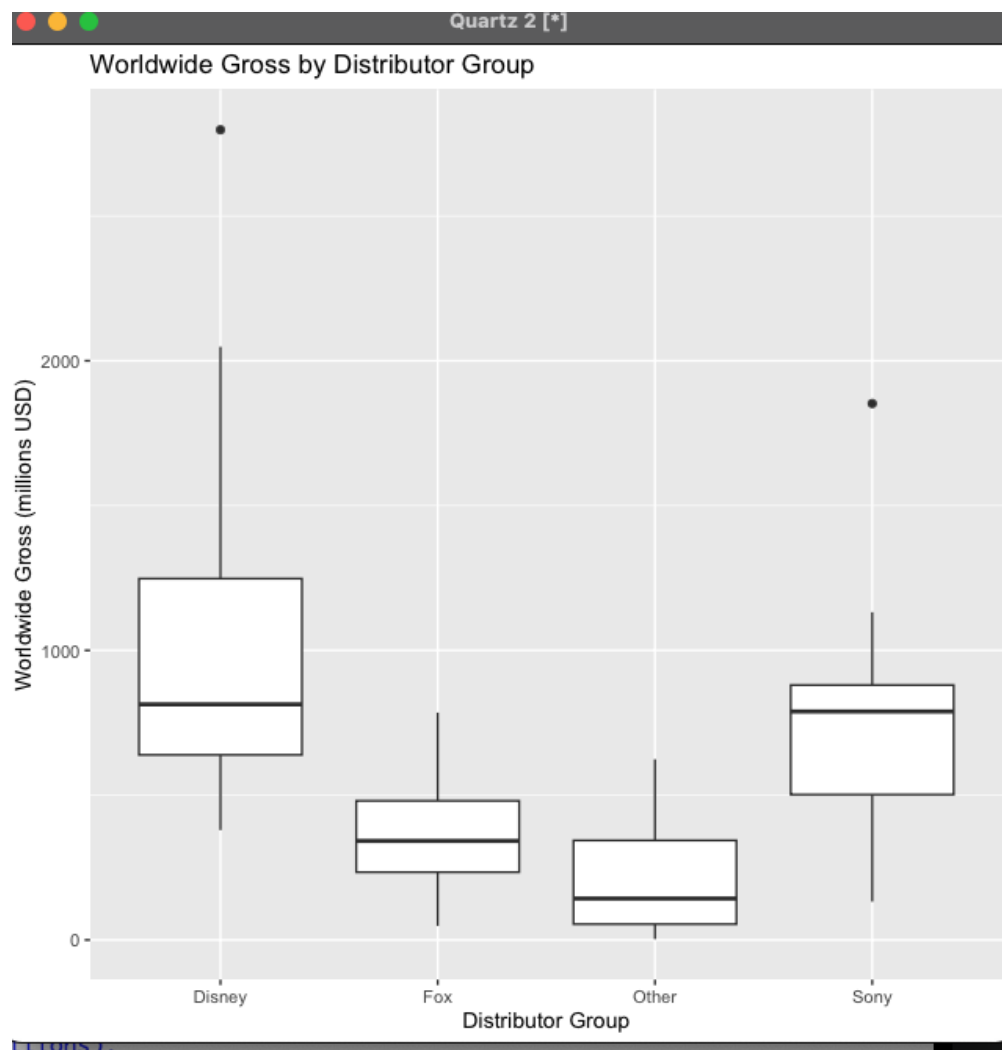


Figure 6: Boxplot of Worldwide Gross by Distributor Group

RQ3: How do budget, Rotten Tomatoes score, and release year predict worldwide gross?

We fit a multiple linear regression model using log-worldwide gross. Multiple linear regression is appropriate here because it allows us to examine how several predictors—budget, Rotten Tomatoes score, and release year—jointly explain variation in worldwide revenue while controlling for each other.

$$\log_{10}(\text{Worldwide Gross}) = \beta_0 + \beta_1(\log_{10} \text{Budget}) + \beta_2(\text{RT Score}) + \beta_3(\text{Year})$$

Regression Output

- Budget_log: $\beta = 1.1037$, $p < 10^{-10}$ (very strong predictor)

- Rotten Tomatoes: $\beta = 0.0059$, $p = 2.6 \times 10^{-6}$ (higher ratings $\rightarrow$ higher revenue)
- Year: not statistically significant
- $R^2 = 0.793$ (model explains 79% of variability)

Prediction Example

Movie with:

- Budget = \$150M
- Rotten Tomatoes = 80
- Year = 2025

Predicted worldwide revenue:

$$\hat{Y} = \$684 \text{ million}$$

Prediction interval:

$$[\$251 \text{ M}, \$1.87 \text{ B}]$$

Interpretation: Even with similar budgets and ratings, movie outcomes vary widely.

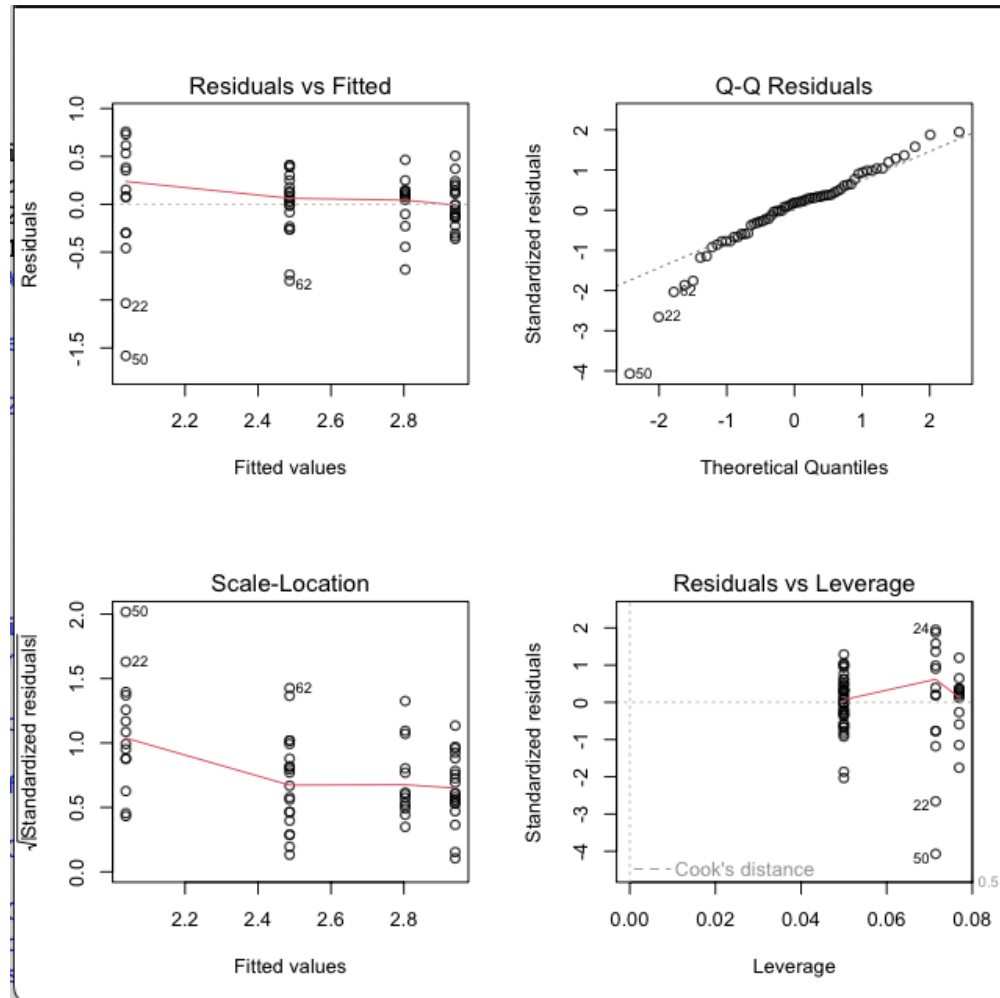


Figure 7: Regression Diagnostic Plots

Conclusion

This project examined financial and critical factors associated with Marvel movie success. Several important conclusions emerged:

- Movies released after 2008 earn significantly more, confirming the impact of the MCU era.
- Disney-distributed movies outperform those from other studios.
- A movie's budget and Rotten Tomatoes score are both strong predictors of worldwide revenue.
- Release year, after accounting for budget and ratings, does not significantly affect worldwide gross.

- Since the dataset contains the full population of Marvel films, these results describe this universe of movies rather than generalizing to all films.

Overall, the combination of strong financial backing and positive audience reception plays a major role in determining box office success.